

Soham Lodh

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SUMMARY

B.Tech CSE (AI & ML) undergraduate at KIIT (CGPA: 9.29) with hands-on experience building production-deployed ML pipelines and full-stack MERN/Next.js applications. Delivered end-to-end systems from data cleaning and model optimization to secure REST APIs, cloud deployment, and role-based access control.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C

ML / Data: XGBoost, Random Forest, Scikit-learn, pandas, NumPy, Matplotlib, Seaborn, statsmodels, Feature Engineering, Hyperparameter Tuning, VIF Analysis

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Middleware, Prisma ORM

Frontend: React.js, Next.js, Tailwind CSS, Axios

Databases: MongoDB, PostgreSQL

Tools & Platforms: Git, GitHub, Vercel, Render, Cloudinary, Streamlit, Postman

EXPERIENCE

Technical Member – Web Development

Aug 2025 – Present

Coding Ninjas KIIT Chapter

Bhubaneswar, India

- Engineered a **full-stack event management platform** (Next.js + Hono + Prisma + MongoDB) enabling participant registration, JWT-authenticated judge workflows, live audience voting, and real-time leaderboard aggregating judge and audience scores simultaneously.
- Implemented a secure **quiz platform** with tab-switch detection, session tracking, and timed execution, dynamically serving randomized question sets from a MongoDB-backed question bank.
- Designed and developed a **Project Management System** (Next.js + NextAuth + MongoDB) supporting Admin, Domain Lead, Project Lead, and Member roles with project lifecycle management, membership workflows, and announcement systems.
- Built scalable REST APIs and responsive interfaces supporting concurrent multi-user interactions across all platforms.

Open Source Contributor

Jul 2025 – Oct 2025

GirlScript Summer of Code

Remote

- Engineered secure authentication modules integrating OTP login, JWT sessions, Google OAuth, and password recovery workflows using React and Express.
- Diagnosed and resolved production-level issues; contributed backend improvements adopted into the main codebase.
- Collaborated with distributed contributors using Git branching workflows and pull request reviews.

General Volunteer

Sep 2025 – Present

NSS, KIIT

Bhubaneswar, India

- Coordinated community initiatives requiring structured planning, execution, and team collaboration.

PROJECTS

PremiumIQ – Health Insurance Premium Predictor

Apr 2026

XGBoost · FastAPI · React

Live: <https://insurance-premium-predictor-drab.vercel.app/>

GitHub: <https://github.com/Soham-Lodh/Insurance-Premium-Predictor>

- Constructed end-to-end ML pipeline on 9,905 records; engineered a `total_risk_score` feature from compound medical history fields, replacing high-cardinality categoricals and improving predictive signal.
- Designed hybrid encoding pipeline (ordinal + one-hot) and validated all 18 features using VIF analysis (all VIF < 2.5), ensuring low multicollinearity.
- Optimized XGBoost via `RandomizedSearchCV` (3-fold CV, 15 iterations); achieved $R^2 = 0.9944$, outperforming Linear Regression (0.956) and tuned Random Forest (0.978).
- Built production inference pipeline (FastAPI + Uvicorn) and integrated React frontend delivering real-time predictions with confidence intervals and risk factor breakdown.

Prescripto – Medical Appointment Platform

Oct 2025 – Jan 2026

MERN Stack · JWT · Cloudinary

Live (User): <https://prescripto-o5lf.vercel.app/>

Live (Admin): <https://prescripto-admin-lyart.vercel.app/>

GitHub: <https://github.com/Soham-Lodh/Prescripto>

- Architected two independently deployed panels (user frontend, admin panel) backed by a unified Express REST API across 20+ endpoints and a single MongoDB Atlas database.
- Implemented role-specific JWT authentication (patient, admin) with bcrypt hashing, input sanitization using `validator.js`, and API rate limiting for security.
- Developed real-time slot booking system preventing race conditions, enabling automatic slot release on cancellation.
- Integrated Cloudinary-based media storage via Multer memory streams and implemented dynamic sitemap generation for improved SEO and discoverability.

EDUCATION

KIIT University

2024 – 2028

B.Tech, Computer Science Engineering (AI & ML)

CGPA: 9.29

Aditya Academy Senior Secondary

2024

Class XII (CBSE)

85%

Aditya Academy

2022

Class X (CBSE)

85.66%

CERTIFICATIONS

- TensorFlow on Google Cloud – Google (2025)
- DSA in Python – NPTEL (2025)
- Postman API Student Expert – Postman (2025)
- GirlScript Summer of Code – Open Source Contributor (2025)

ADDITIONAL INFORMATION

Languages: English (Fluent), Bengali (Native), Hindi (Fluent)

Interests: ML Systems, Backend Engineering, Scalable Web Applications